SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I V. THANISHA / 196526016D *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: TV

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Concepts	25%	2	2	2	2
Explanation	25%	2	2	2	2
Application	20%	2	2	2	2
Organization	15%	2	1	1	1
Accuracy	10%	1	1	1	1
Clarity	5%	1	-	1	-

Total Marks 34 **/40**

Signature of the Faculty

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7.An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests

of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9.A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10.A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation

Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organizatio n	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentatio n	Understanda ble with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

CO1 - ASSESSMENT - 1

NAME: V Thanisha
COURSE: Computer Network
CODE: CSAD709
REG NO: 196526O160
DEPT: CSC (Cyber)

1. GIVEN:

FILE SIZE = 150 MB
BANDWIDTH = 56 Mbps
SEGMENT SIZE = 1500 BYTES
DATA LINK OVERHEAD PER FRAME = 40 BYTES

34/40

As,

1 MB = 1,000,000 Bytes
1 Byte = 8 Bits
150 MB = 150 × 1,000,000
= 150,000,000 Bytes

Number of Bytes = File Size / Segment Size
= 150,000,000 / 1500
= 1000,000 Segments.

No. of frames = 1000,000
Each frame has 40 Bytes
Total overhead = 1000,000 × 40
= 4000000 Bytes
= 4 MB

Total Data link overhead = 4 MB.

Original Data = 150 MB.

Overhead = 4 MB.

Total Transmission ~~time~~ data = 150 + 4

= 154 MB.

Converting to Bits = 154 × 8 = 1232

Transmission Time = $\frac{1232}{50}$

= 24.64 Seconds

No. of Segments = 100,000

No. of frames = 100,000

Data link overhead = 4 MB

Total transmitted data = 154 MB

Transmission Time = 24.64 Seconds.

GIVEN:

WINDOW SIZE = 6 FRAMES
EACH FRAME = 1024 Bytes.
TOTAL FRAMES = 48 FRAMES.
LOST FRAME PER WINDOW = 1

Number of windows = $\frac{\text{Total Frames}}{\text{Window Size}}$

$= \frac{48}{6}$

$= 8$

One Frame is lost in every window
There are 8 windows

Retransmission = 8×1
$= 8$

Total Retransmission = 8 Frames

* Multiple frames can be transmitted before waiting for acknowledgements
* The receiver permits an appropriate amount of outstanding data.

* Network Utilization improves
* Waiting time is reduced.
* The sender does not Unecessarily overloaded the receiver

ROLE OF FLOW CONTROL:

Flow control prevents the sender from transmitting data faster than the receiver can process.

Therefore,

* Transmission window = 8
* Total Retransmission = 8
* Flow control improve efficiency by allowing multiple frames to be sent before acknowledgement while preventing receiver overload.

GIVEN:

* IP PACKET = 12,000 Bytes.
* MTU = 1500 Bytes
* IP Header per Fragment = 20 Bytes.

Maximum fragment size = 1500 Bytes

Header = 20 Bytes.

Therefore,

Data per full fragment = 1500 - 20
= 1480 Bytes

Number of Fragments = $\frac{12000}{1480}$
= 8.108

Therefore,

8 complete fragments are not enough

Number of fragments = 9

The First 8 Fragment carries

= 8 × 1480
= 11840 Bytes.

Remaining data,

12000 - 11840 = 160 Bytes

So, the 9th fragment carries 160 Bytes of data plus its 20-Byte header

Each Fragment has a 20-Byte header

= 9 x 20
= 180 Bytes

Therefore,

Total IP header overhead = 180 Bytes.

* Fragments = 9
* Full Fragments = 8
* Final Fragment carries = 160 Bytes
* Total IP header ~~hea~~ overhead = 180 Bytes.

How successful delivery is Achieved.

The network layer ensures proper reassembly using:

1. IDENTIFICATION FIELD:

All fragments belonging to the same original IP packet have the same identification value.

2. FRAGMENT OFFSET:

The offset indicates the position of each fragment with the original packet

3. MORE FRAGMENTS FLAG:

This indicates whether additional fragment follows.

4. DESTINATION REASSEMBLY:

The destination collects the fragments and reconstruct the original packet.

5. GIVEN:

Total file = 600 MB

Checkpoint interval = 60 MB

Failure occurs after 390 MB.

Checkpoints occurs at,

* 60 MB
* 120 MB
* 180 MB
* 240 MB
* 300 MB
* 360 MB.

The next checkpoint would be at 420 MB

But failure occurs at 390 MB

Therefore,

6 checkpoints

The most recent checkpoint was at 360 MB

failure occured at,

390 MB.

IMPORTANCE OF SESSION LAYER SYNCHRONIZATION

synchronization checkpoints allows communication to resume from a known point rather than starting from the beginning

Without checkpoints,

390 MB

might have to be retransmitted.

With the checkpoint,

30 MB

needs to be retransmitted

This saves:

390 - 30 = 360 MB

of Unnecessary retransmission.